Research and Literature Review

Research Background

Bitcoin has become an important subject in financial research because its returns may be associated with developments in traditional financial markets and the broader economy. Although Bitcoin operates independently of conventional banking systems, its market performance may coincide with changes in investor confidence, interest rates, exchange rates, and commodity prices. Examining these relationships can help researchers understand whether Bitcoin responds to the same financial conditions that influence other assets.

This study examines the relationship between Bitcoin's monthly log returns and eight financial and economic indicators from January 2018 to December 2025. These include S&P 500 returns, U.S. Dollar Index (DXY) returns, changes in the U.S. 10-year Treasury yield, the CBOE Volatility Index (VIX), Philippine M3 growth, changes in the Bangko Sentral ng Pilipinas (BSP) overnight reverse repurchase (RRP) rate, changes in the USD/PHP exchange rate, and West Texas Intermediate (WTI) crude oil returns.

Review of Related Literature

Panagiotidis, Stengos, and Vravosinos (2019) examined how financial markets, uncertainty, and public interest relate to Bitcoin returns. Using vector autoregression (VAR) and factor-augmented VAR (FAVAR) models, they considered stock-market returns, exchange rates, gold and oil returns, central-bank interest rates, and internet search activity. Their findings showed significant interactions between Bitcoin and traditional stock markets, while interactions with foreign-exchange markets and the broader macroeconomy were weaker. This suggests that traditional financial-market conditions are worth considering when examining Bitcoin returns, although their influence may differ across variables.

Jareño et al. (2020) studied Bitcoin's relationship with international financial risk factors, including U.S. stock-market returns, interest rates, crude oil prices, and the VIX. Using quantile regression and a nonlinear autoregressive distributed lag (NARDL) approach, the researchers found that the VIX was an important risk factor, with negative and statistically significant effects on Bitcoin returns across most of the periods and quantiles examined. U.S. stock-market returns had positive and statistically significant effects in their model, while oil returns had negative effects at lower Bitcoin-return quantiles. The study also found that Bitcoin's sensitivity to international risk factors was more pronounced under extreme market conditions. These findings are relevant to the present study because several of its explanatory variables overlap with those examined by Jareño et al. However, their findings cannot automatically be assumed to apply to the 2018–2025 sample.

Monetary policy is another factor that may be associated with Bitcoin's performance. Karau (2023) examined the effects of U.S. monetary policy and found that Bitcoin's response to policy changes evolved over time. Since late 2020, Bitcoin prices have tended to fall immediately following Federal Reserve tightening announcements, whereas earlier in its history, tightening had been associated with increases in Bitcoin prices. This suggests that Bitcoin's relationship with monetary conditions can change over time. Although the study focuses on monetary-policy announcements rather than changes in the U.S. 10-year Treasury yield specifically, it provides a basis for investigating whether interest-rate movements are associated with Bitcoin's monthly returns.

Exchange rates and Philippine monetary conditions provide additional areas for investigation. Panagiotidis et al. (2019) included exchange rates in their analysis but found weaker interactions between Bitcoin and foreign-exchange markets than between Bitcoin and traditional stock markets. The present study examines both DXY returns and changes in the USD/PHP exchange rate. These indicators measure different aspects of currency movements: DXY tracks the U.S. dollar against a basket of major currencies, while USD/PHP measures the peso's value against the dollar. The study also includes Philippine M3 growth and changes in the BSP overnight RRP rate to examine whether domestic monetary conditions are associated with Bitcoin returns. Their inclusion extends the analysis to Philippine economic conditions, but their relationships with Bitcoin returns remain to be tested.

WTI crude oil returns are included because oil prices reflect changes in global commodity markets and may move alongside broader economic and financial conditions. Jareño et al. (2020) found that the relationship between oil returns and Bitcoin differed across Bitcoin-return quantiles. This suggests that the relationship may depend on market conditions rather than follow a consistent direction. The present study will therefore test whether WTI returns are associated with Bitcoin's monthly log returns during the selected period.

Research Gap

Previous studies have examined Bitcoin's relationships with stock-market performance, market volatility, monetary policy, exchange rates, and commodity prices. Their findings suggest that these relationships can differ across periods and market conditions. However, the studies reviewed do not directly establish whether Philippine M3 growth and changes in the BSP overnight RRP rate are associated with Bitcoin returns when examined alongside international financial indicators.

This study extends previous research by examining whether global financial-market indicators and Philippine monetary variables are jointly associated with Bitcoin's monthly log returns during the January 2018–December 2025 period. Using multiple regression, it will assess whether the eight explanatory variables have statistically significant associations with Bitcoin's monthly log returns. The results will help identify which variables, if any, are associated with Bitcoin returns in the selected sample. These relationships will be interpreted as statistical associations rather than proof of causation.

References

Jareño, F., González, M. de la O., Tolentino, M., & Sierra, K. (2020). Bitcoin and gold price returns: A quantile regression and NARDL analysis. *Resources Policy, 67*, Article 101666. https://doi.org/10.1016/j.resourpol.2020.101666

Karau, S. (2023). Monetary policy and Bitcoin. *Journal of International Money and Finance, 137*, Article 102880. https://doi.org/10.1016/j.jimonfin.2023.102880

Panagiotidis, T., Stengos, T., & Vravosinos, O. (2019). The effects of markets, uncertainty and search intensity on bitcoin returns. *International Review of Financial Analysis, 63*, 220–242. https://doi.org/10.1016/j.irfa.2018.11.002